It is recommended that you spend about 30 minutes on this question.

2 Many homes have smoke detectors fitted to the ceilings of certain rooms to provide an early warning of a fire. These detectors contain a weak radioactive source that ionises the air between two metal plates. See Fig. 2.1.

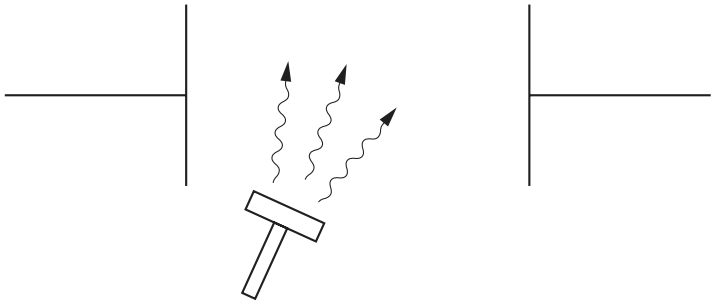

Fig. 2.1

A low voltage battery in the detector causes the ions to move. This produces a very small ionisation current in a circuit containing the battery and the plates. Any reduction in this ionisation current due to smoke is detected and an alarm sounds.

An airline company wishes to install some of these smoke detectors in its aircraft and needs to know if changes in air pressure will affect the ionisation current.

Design a laboratory experiment to investigate how the ionisation current depends on air pressure. You should draw a detailed labelled diagram showing the arrangement of your apparatus. In your account you should pay particular attention to

(a) the type of source used in the investigation (i.e. whether it is an alpha, beta or gamma emitter) giving a reason for your choice,

(b) how the ionisation current would be measured (given that it is very small),

(c) the method of changing and measuring the air pressure,

(d) the procedure to be followed,

(e) any safety precautions that you would take.

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